Lecture 3:
Sample-Based Classification

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10 ECTS Credits

Department of Information and Communication Technology

Faculty of Engineering and Science

UiA

Recap

- Classification without features
- Classification with features
- Classifiers and Classification Error
- Bayes Classifier
- Bayes Error
- Bayes Error for a Transformed Feature Vector
- Minimax Classification
- Bayes' Decision Theory
- Special Case of Bayes Decision Theory

Classification Rules

- Optimal classification requires full knowledge of the stochastic relationship between $\boldsymbol{X}$ and $Y, P_{\boldsymbol{X},Y}$.
- We employ a mixture of distributional knowledge and **training data**, given by:

$$S_n = \{(\boldsymbol{X}_1, Y_1), (\boldsymbol{X}_2, Y_2), \dots, (\boldsymbol{X}_n, Y_n)\}$$

- We assume the training data S_n is **independent and identically distributed (i.i.d.)** sample from $P_{\boldsymbol{X},Y}$:
  - Each data point is sourced independently from the same foundational probability distribution.
  - Every data instance possesses an equivalent probability of occurrence, and the value of a single data point does not influence the value of another.
- S_n: Comes from performing a vector measurement $\boldsymbol{X}_i$ on each of the n specimens and an expert producing a label $Y_i = 0$ with $P(\,Y = 0 \mid \boldsymbol{X}_i\,)$ and a label $Y_i = 0$ with $P(\,Y = 0 \mid \boldsymbol{X}_i\,)$.

Classification Rules

- The class-specific sample sizes:

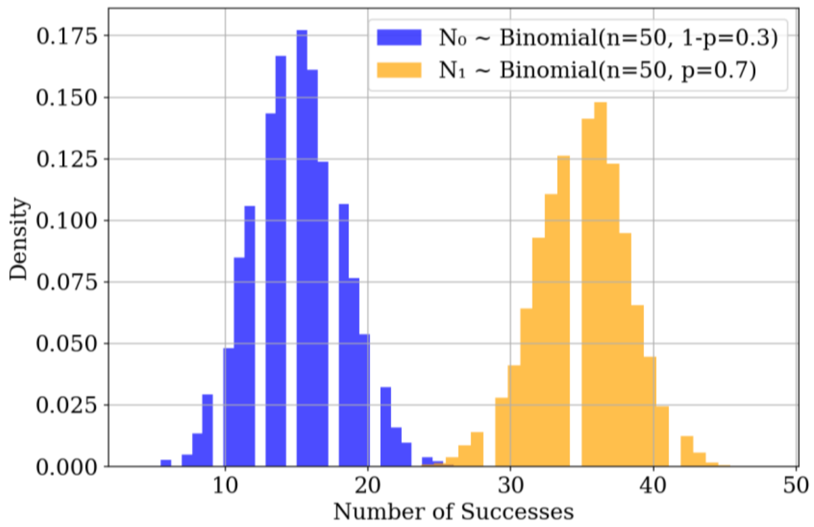
$$N_0 = \sum_{i=1}^{n} I_{Y_i=0} \text{ and}$$

$$N_1 = \sum_{i=1}^{n} I_{Y_i=1}, \text{ s.t. } n = N_0 + N_1$$

  are binomial random variables with parameters $(n, 1-p)$ and (n, p), respectively, where $p = P(Y = 1)$.
- A **classification rule** is an operator that outputs a trained classifier ψ_n. Former outputs classifiers while latter outputs class labels.
- If $\mathcal{C}$ denotes the set of all classifiers of interest:

$$\Psi_n: [\mathbb{R}^d \times \{0,1\}]^n \to \mathcal{C}$$

- In other words, a classification rule, Ψ_n maps sample data $S_n \in [\mathbb{R}^d \times \{0,1\}]^n$ into a classifier $\psi_n = \Psi(S_n) \in \mathcal{C}$.



Mean, μ: $E(X) = np$

Variance, σ^2: $np(1-p)$

Nearest-Centroid Classification Rule

- Consider a classifier:

$$\psi_n(\boldsymbol{x}) = \begin{cases} 1, & \|\boldsymbol{x} - \hat{\boldsymbol{\mu}}_1\| < \|\boldsymbol{x} - \hat{\boldsymbol{\mu}}_0\|, \\ 0, & \text{otherwise}, \end{cases}$$

  where

$$\hat{\boldsymbol{\mu}}_0 = \frac{1}{N_0}\sum_{i=1}^{n} \boldsymbol{X}_i I_{Y_i=0} \text{ and } \hat{\boldsymbol{\mu}}_1 = \frac{1}{N_1}\sum_{i=1}^{n} \boldsymbol{X}_i I_{Y_i=1}$$

  are the sample means (centroids) for each class ⇒ the classifier assigns the test point $\boldsymbol{x}$ the label of the nearest class mean.
- Thus, the classification rule produces hyperplane decision boundaries.
- By replacing sample mean, with other kinds of centroids (e.g. sample median), a family of like-minded classification rules can be obtained.

Nearest-Centroid Classification Rule

- Nearest-Centroid Classification Rule:

$$\psi_n(\boldsymbol{x}) = \begin{cases} 1, & \|\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_1\| < \|\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_0\|, \\ 0, & \text{otherwise.} \end{cases}$$

- The squared Euclidean distance $\left(d = \sqrt{(\boldsymbol{x}_2 - \boldsymbol{x}_1)^2 + (y_2 - y_1)^2}\right)$ is given by:

$$\|\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_i\|^2 = (\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_i)^T(\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_i) = \boldsymbol{x}^T\boldsymbol{x} - 2\boldsymbol{x}^T\widehat{\boldsymbol{\mu}}_i + \widehat{\boldsymbol{\mu}}_i{}^T\widehat{\boldsymbol{\mu}}_i$$

$$\|\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_1\| < \|\boldsymbol{x} - \widehat{\boldsymbol{\mu}}_0\| \Leftrightarrow -2\boldsymbol{x}^T\widehat{\boldsymbol{\mu}}_1 + \widehat{\boldsymbol{\mu}}_1{}^T\widehat{\boldsymbol{\mu}}_1 < -2\boldsymbol{x}^T\widehat{\boldsymbol{\mu}}_0 + \widehat{\boldsymbol{\mu}}_0{}^T\widehat{\boldsymbol{\mu}}_0$$

- Rearranging,

$$2\boldsymbol{x}^T(\widehat{\boldsymbol{\mu}}_1 - \widehat{\boldsymbol{\mu}}_0) > (\widehat{\boldsymbol{\mu}}_1{}^T\widehat{\boldsymbol{\mu}}_1 - \widehat{\boldsymbol{\mu}}_0{}^T\widehat{\boldsymbol{\mu}}_0)$$

- Substituting $\boldsymbol{a}_n = \widehat{\boldsymbol{\mu}}_1 - \widehat{\boldsymbol{\mu}}_0$ and $b_n = (\widehat{\boldsymbol{\mu}}_1 - \widehat{\boldsymbol{\mu}}_0)(\widehat{\boldsymbol{\mu}}_1 + \widehat{\boldsymbol{\mu}}_0)^T/2$
- The inequality becomes:

$$\boldsymbol{a}_n^T\boldsymbol{x} + b_n > 0$$

- Nearest-Centroid Classification Rule could be rewritten as:

$$\psi_n(\boldsymbol{x}) = \begin{cases} 1, & \boldsymbol{a}_n^T\boldsymbol{x} + b_n > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Nearest-Neighbor Classification Rule

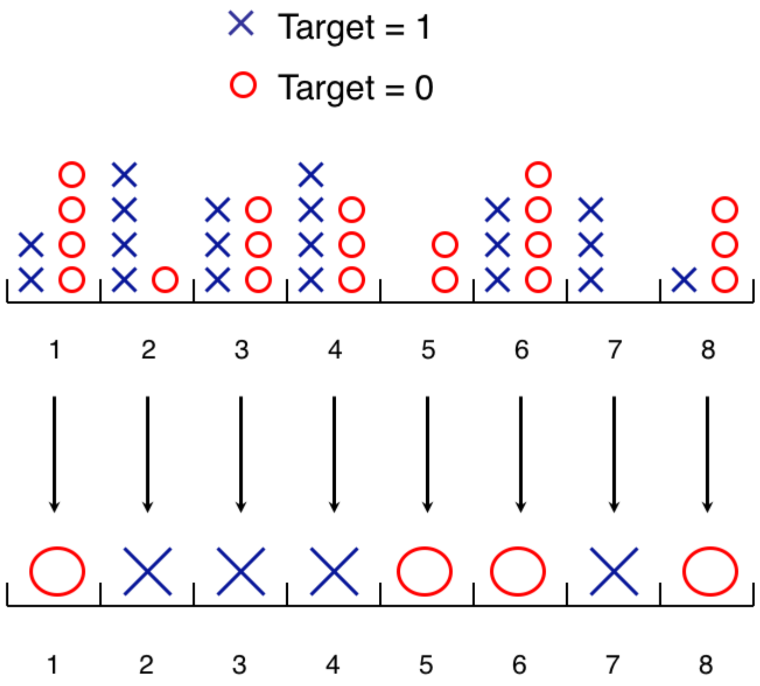
- Consider a classifier:
$$\psi(\boldsymbol{x}) = Y_{(1)}(\boldsymbol{x}),$$
where $\left(\boldsymbol{X}_{(1)}(\boldsymbol{x}), Y_{(1)}(\boldsymbol{x})\right)$ is the nearest training point:
$$\boldsymbol{X}_{(1)}(\boldsymbol{x}) = \arg \min_{\boldsymbol{X}_1,\ldots,\boldsymbol{X}_n} ||\boldsymbol{X} - \boldsymbol{x}||$$
- Classifier assigns the test point $\boldsymbol{x}$ the label of the nearest neighbor in the training data (taking the point with the smallest index i, in case of a tie: an arbitrary rule).
- A family of similar classification rules can be obtained by replacing the Euclidean distance with other metrics (e.g. correlation).
- Generalizing, a classifier could assign test point $\boldsymbol{x}$ the label of the k nearest training points, with odd value for k to avoid ties. ($k = 1$ yields the previous).
- This is called the **$\boldsymbol{k}$-nearest neighbor classification rule**.

Discrete Histogram Classification Rule

- Suppose the input space $\boldsymbol{X}$ consists if a finite number of points $\{\boldsymbol{x}^1, \boldsymbol{x}^2, \dots \boldsymbol{x}^b\} \in R^d$ s.t. each $\boldsymbol{x}^j$ represents a bin/category.
- U_j and V_j are the number of training points where $(\boldsymbol{X}_i = \boldsymbol{x}^j, Y_i = 0)$ and $(\boldsymbol{X}_i = \boldsymbol{x}^j, Y_i = 1)$, respectively $\forall\, j = 1, \dots, b$.
- The discrete histogram rule is given by:

$$\psi_n(\boldsymbol{x}^j) = \begin{cases} 1, & U_j < V_j, \\ 0, & \text{otherwise}, \end{cases} \quad \forall\, j = 1, \dots, b.$$

- The discrete histogram rule assigns to $\boldsymbol{x}^j$ the majority label among the training points that coincide with $\boldsymbol{x}^j$. In the case of a tie, the label is set to zero.



Classification Error Rates

- Given a classification rule Ψ_{n}, the error rate of a designed classifier $\psi_n = \Psi(S_n)$ is given by:

$$\varepsilon_n = P(\,\Psi_n(S_n)(X) \neq Y \mid S_n\,) = P(\,\psi_n(X) \neq Y \mid S_n\,)$$

- (X, Y) is a test point independent of S_n (training data).
- The error ε_n is a function of random sample data S_n and hence a random variable.
- Assuming Ψ_{n} is non-random, on a specific and fixed S_n, ε_n becomes an ordinary real number and is sometimes called the **conditional error**, as it is conditional on the data.

Classification Error Rates

- Another important error rate in sample-based classification is the **expected error** (sometimes called **unconditional error**):

$$\mu_n = E[\varepsilon_n] = P(\psi_n(X) \neq Y)$$

- Among the error rates ε_n is more interesting in practice, because it is the error of classifier designed on a sample data at hand.
- μ_n is independent of data and is a function of the classification rule.
- The most common criterion to compare classification rules is to pick the one with least expected error μ_n, for a fixed given sample size n.
- Class specific error rates:

$$\varepsilon_0 = P(\psi_n(X) = 1 \mid Y = 0, S_n)$$
$$\varepsilon_1 = P(\psi_n(X) = 0 \mid Y = 1, S_n)$$

- Classification Error:

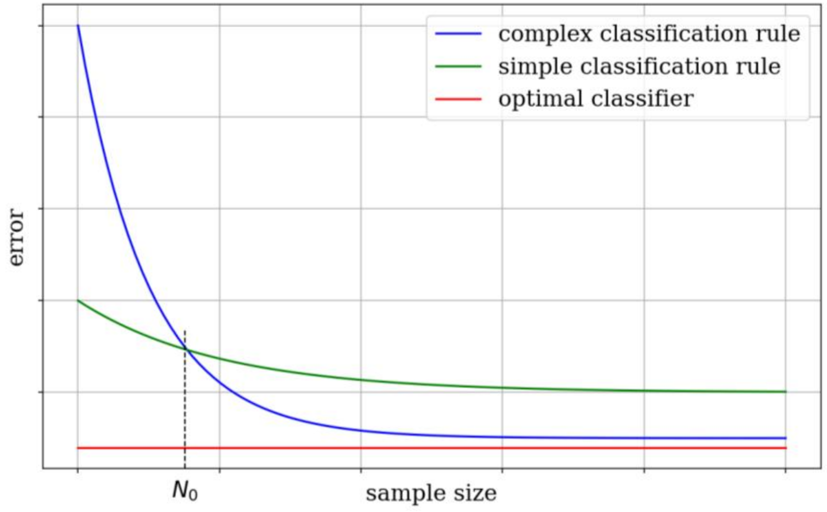
$$\begin{aligned}\varepsilon[\psi_n] &= P(\psi_n(X) \neq Y \mid S_n) \\ &= P(\psi_n(X) = 1 \mid Y = 0, S_n)P(Y = 0) + P(\psi_n(X) = 0 \mid Y = 1, S_n)P(Y = 1) \\ &= P(Y = 0)\varepsilon_n^0[\psi] + P(Y = 1)\varepsilon_n^1[\psi]\end{aligned}$$

Consistency

- Associated with the requirement as $n \to \infty$, the classification error should approach the optimal error.
- Thus as $n \to \infty$, the classification rule Ψ_n is said to be consistent if $\varepsilon_n \to \varepsilon^*$ in probability.
  - For any small positive value $\tau > 0$, the probability that ε_n deviates from ε^* becomes smaller as $n \to \infty$ (aka. **weak/ordinary consistency**):
    $$P(|\varepsilon_n - \varepsilon^*| > \tau) \to 0$$
- A classification rule is **strongly consistent**, if as $n \to \infty$, $P(\varepsilon_n \to \varepsilon^*) = 1$ or $\varepsilon_n \to \varepsilon^*$ with probability 1.
- Strong consistency demands that for all possible sequences of training data S_n, $\varepsilon_n \to \varepsilon^*$.
- Ordinary consistency is enough practically.

Consistency

- A classification rule Ψ_n could be consistent under a feature label distribution, $P_{\boldsymbol{X},Y}$, but not under another.
- A **universally consistent** classification rule is consistent under any distribution.
- Universal consistency is a property of the classification rule alone.
- Universally consistent rules produce complex classifiers, which could result in a “scissors effect”.
- Consistency is a large sample property, and could be irrelevant in small-sample cases, as non-consistent classification rules are typically better than consistent classification rules when the training data size is small.

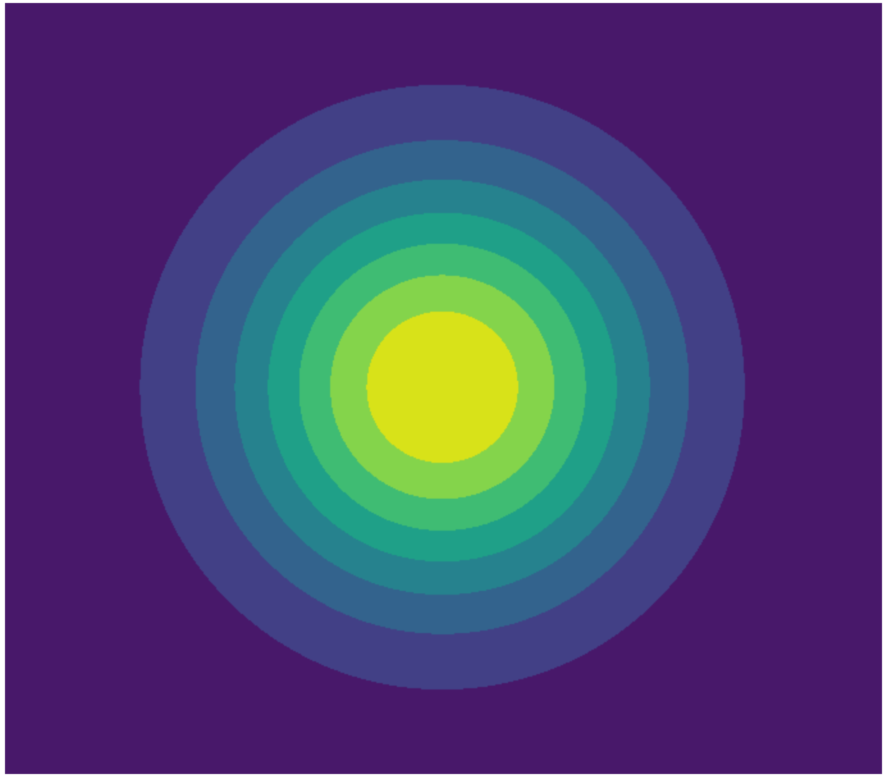


Uniform Bounded Convergence Theorem

Theorem: Let $\{X_n; n = 1,2,\dots\}$ be a uniformly bounded random sequence. The following statements are equivalent:

1. $X_n \to X \ in\ L^p, for\ some\ p > 0$
2. $X_n \to X \ in\ L^q, for\ all\ q > 0$
3. $X_n \to X \ in\ probability$.

- L^p and L^q measure "how close" (average distance) $X_n\ and\ X$ are: $|X_n - X|^p \to 0$.
- 3: The probability that X_n deviates X from by more than a tiny amount ϵ goes to 0.



Uniform Bounded Convergence Theorem

- $\{X_n; n = 1,2, \dots\}$, a sequence of random variables,
- X the limit random variable the sequence converges to.
- Uniform boundedness $|X_n| \leq M$.
- The three convergence are equivalent:
  - L^p convergence:
    $$X_n \to X \text{ in } L^p \text{ means } E[|X_n \to X|^p] \to 0 \text{ as } n \to \infty \text{ for some } p > 0$$
  - L^q convergence:
    $$X_n \to X \text{ in } L^q \text{ means } E[|X_n \to X|^q] \to 0 \text{ as } n \to \infty \text{ for all } q > 0$$
  - Convergence in probability:
    $$X_n \to X \text{ in } P \text{ means } P(|X_n \to X| > \tau) \to 0 \text{ as } n \to \infty \text{ for any } \tau > 0$$

Consistency in Terms of the Expected Error

- **Theorem**: The classification rule Ψ_n is consistent iff., as $n \to \infty$,

$$E[\varepsilon_n] \to \varepsilon^*$$

Proof:

- $\{\varepsilon_n : n = 1,2, \dots\}$ is a uniformly bounded random sequence, as $0 \leq \varepsilon_n \leq 1 \; \forall \, n$
- By Uniform Bounded Convergence Theorem,

$$\text{Convergence in probability } (\varepsilon_n \to \varepsilon^*) \Leftrightarrow L^1 \text{ convergence } (E[|\varepsilon_n - \varepsilon^*|] \to 0)$$

- Also, $\varepsilon_n \geq \varepsilon^* \Rightarrow |\varepsilon_n - \varepsilon^*| = \varepsilon_n - \varepsilon^*$(Bayes error is the theoretically possible least error)

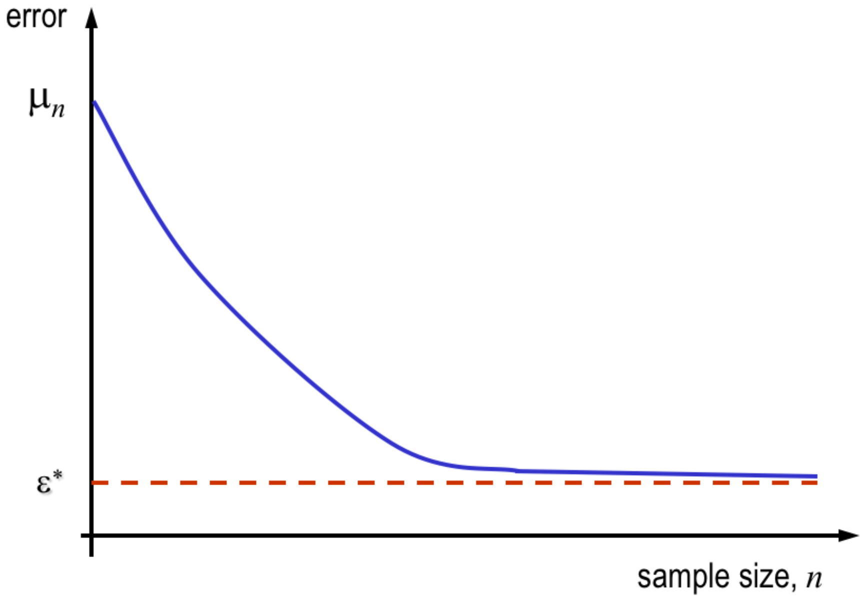
$$E[|\varepsilon_n - \varepsilon^*|] = E[\varepsilon_n] - \varepsilon^*$$

- Thus,

$$E[|\varepsilon_n - \varepsilon^*|] \to 0 \Rightarrow E[\varepsilon_n] - \varepsilon^* \to 0 \Rightarrow E[\varepsilon_n] \to \varepsilon^*$$

Consistency in Terms of the Expected Error

- Theorem demonstrates that consistency is characterized entirely by the first moment (expectation/mean) of the random variable ε_n as n increases.
- Weak consistency is gauranteed if the average error ($\mu_n = E[\varepsilon_n]$) converges to ε^*.
- Strong consistency requires $\varepsilon_n \rightarrow \varepsilon^*$ almost surely (entire distribution).
- Weak consistency can be verified by plotting μ_n against sample size as shown here.



Consistency of Nearest Neighbor Rule

- The expected error of the nearest neighbor classification rule satisfies
$$\lim_{n\to\infty} E[\varepsilon_n] \le 2\varepsilon^*$$
- Assume that the feature-label distribution is such that it is possible for a perfect classifier to exist: $\varepsilon^* = 0$
- Thus,
$$\lim_{n\to\infty} E[\varepsilon_n] \le 0$$
- Since error rates are non-negative, $\lim_{n\to\infty} E[\varepsilon_n] = 0$
- By the previous theorem, the NN classification rule is consistent in this scenario.

No-Free-Lunch Theorems

- Universal consistency implies that no knowledge of the feature-label distribution is necessary to obtain optimal performance model.
- If one has access to large enough sample, purely data-driven approach obtains performance arbitrarily close to optimal performance.
- The No-Free-Lunch Theorems show that this is deceptive.
- Some knowledge about the feature-label distribution must be obtained to guarantee acceptable performance.
- Proofs available in (Theorem 7.1 and 7.2): Devroye, Luc, László Györfi, and Gábor Lugosi. *A probabilistic theory of pattern recognition*. Vol. 31. Springer Science & Business Media, 2013.

No-Free-Lunch Theorems

- **Theorem**: For every $\tau > 0$, integer n, and classification rule Ψ_n, there exists a feature-label distribution $P_{\mathbf{X},Y}$ (with $\varepsilon^* = 0$) such that $E[\varepsilon_n] \geq \frac{1}{2} - \tau$.
- The first theorem states that **no classifier is universally reliable** at finite sample sizes.
- Even if the classes are perfectly separable, there exists a "worst-case" distribution where the classifier error is arbitrarily close to error in random guessing.
- In the case of universally consistent classification rules, this means that no matter how large n is, one can never know if their finite-sample performance will be satisfactory (unless one knows something about the feature-label distribution).

No-Free-Lunch Theorems

- **Theorem**: Given any sequence $\{a_n\}$ (where $a_n \to 0$) and any classification rule Ψ_n, there is a feature-label distribution $P_{\mathbf{X},Y}$ (with $\varepsilon^* = 0$) such that $E[\varepsilon_n] \geq a_n, \forall n = 1,2, \ldots$
- The feature-label distribution in the previous theorem may have to be different for different n.
- The second theorem applies to fixed feature-label distribution and states that though one may get $E[\varepsilon_n] \to \varepsilon^*$, purely data-driven, one must know something about the feature-label distribution in order to guarantee rate of convergence.

NFL Theorems Insights

- While universal consistency ensures optimal performance as $n \to \infty$, NFL theorems show that finite-sample guarantees are impossible without prior knowledge of $P_{\mathbf{X},Y}$.
- All classification rules are equally vulnerable to adversarial distributions. To avoid "worst-case" scenarios, practitioners must incorporate domain knowledge (e.g., assuming linearity or low dimensionality).
- The theorems emphasize that machine learning is not purely algorithmic, it requires understanding the data structure.

Thank You

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